

Late-Onset Pompe Disease (LOPD): Comprehensive Disease Characteristics Report

Disease: Late-Onset Pompe Disease (LOPD) **Synonyms:** Glycogen storage disease type II (GSD-II), acid maltase deficiency (AMD), acid α -1,4-glucosidase deficiency, glycogenosis type II **Category:** Mendelian, autosomal recessive lysosomal storage disorder **Suggested MONDO ID:** MONDO:0009290 (Pompe disease); **OMIM:** 232300; **Gene:** GAA (HGNC:4065, 17q25.3)

Summary

Late-Onset Pompe Disease (LOPD) is an autosomal-recessive lysosomal storage disorder caused by biallelic **partial** loss-of-function variants in the GAA gene, which encodes the lysosomal enzyme acid α -1,4-glucosidase (acid maltase). Residual enzyme deficiency leads to progressive intralysosomal glycogen accumulation, secondary failure of autophagy, and disruption of the contractile apparatus — preferentially in **type II (fast-twitch) skeletal and respiratory muscle fibers**. Unlike the infantile-onset form (IOPD), which is dominated by lethal hypertrophic cardiomyopathy, LOPD presents after the first year of life (childhood through late adulthood) as a **slowly progressive limb-girdle and axial myopathy with disproportionately early diaphragmatic/respiratory insufficiency and minimal cardiac involvement**. Respiratory failure is the leading cause of death.

The molecular hallmark is the "leaky" splice variant **c.-32-13T>G (IVS1)**, which appears in ~90% of Caucasian late-onset patients and permits low-level production of functional enzyme (hence a comparatively mild, adult phenotype and generally CRIM-positive immunological status). Diagnosis rests on demonstrating reduced GAA activity on a **dried blood spot** (fluorometry or FIA-MS/MS), confirmed by GAA sequencing, with urinary **glucose tetrasaccharide (Glc4/Hex4)** and serum creatine kinase (CK) as supporting biomarkers. Pseudodeficiency alleles (notably p.[G576S;E689K], common in East Asians) are an important cause of screening false positives.

Management centers on **enzyme replacement therapy (ERT)**. Three agents are now approved: alglucosidase alfa (>15 years standard of care), avalglucosidase alfa (enhanced bis-M6P mannose-6-phosphate uptake), and cipaglucosidase alfa co-administered with the chaperone miglustat. ERT modifies but does not halt the disease: respiratory and ambulatory function improve initially then decline gradually over 10–15 years. Next-generation ERTs show superiority signals over alglucosidase alfa in walk distance, lung function, biomarker normalization, and patient-reported outcomes. Multidisciplinary supportive care — respiratory support, supervised resistance/respiratory-muscle training, and high-protein nutrition — provides meaningful adjuvant benefit. Gene therapy (AAV-GAA) is in trials but has documented risks (dorsal-root-ganglion sensory neuronopathy).

Section 1 — Disease Information

LOPD is a rare inherited neuromuscular glycogen storage disorder resulting from deficiency of the lysosomal enzyme acid α -glucosidase (GAA), which normally degrades lysosomal glycogen to glucose. When enzyme activity is only partially reduced and symptoms appear after 12 months of age, the condition is termed **late-onset** to distinguish it from the severe classic infantile form.

Key identifiers (suggested):

Resource	Identifier
MONDO	MONDO:0009290 (Pompe disease)
OMIM	232300 (Glycogen storage disease II)
Gene (HGNC)	GAA, HGNC:4065
ICD-10	E74.02 (Pompe disease)
MeSH	D006009 (Glycogen Storage Disease Type II)
Orphanet	ORPHA:308552 (late-onset form) / ORPHA:365 (GSD-II)

Synonyms/alternative names: Glycogen storage disease type II (GSD-II), acid maltase deficiency, acid α -1,4-glucosidase deficiency, glycogenosis type II, adult/juvenile-onset Pompe disease.

Information is derived predominantly from **aggregated disease-level resources** — international patient registries (e.g., the International Pompe Registry, n=485), newborn screening cohorts, and randomized controlled trials — rather than from single-patient EHR data. LOPD is described in the literature as *"a rare autosomal recessive multisystemic disorder"* that substantially impacts daily activities and health-related quality of life ([PMID: 39535661](#)).

Section 2 — Etiology

Primary cause (genetic): LOPD is caused by biallelic pathogenic variants in *GAA* producing **partial** (rather than complete) loss of acid α -glucosidase function. The dominant late-onset allele is the intron-1 leaky splice variant **c.-32-13T>G**, which allows a small fraction of normally spliced transcript and thus residual enzyme. *"This splicing variant occurs in 90% of Caucasian late onset patients, and is associated with a broad range of symptom onset"* ([PMID: 30922962](#)).

Genetic risk/modifier factors: The *cis* synonymous variant **c.510C>T** acts as a genetic modifier — *"c.510C>T is a genetic modifier in compound heterozygous and homozygous IVS1 patients"* — reducing residual wild-type splicing and lowering the age of onset ([PMID: 30922962](#)). Rare variants in autophagy and non-autophagy genes have been proposed as additional disease modifiers explaining variable severity among patients sharing the same *GAA* genotype ([PMID: 33807278](#)).

Environmental/lifestyle risk factors: No established environmental cause; this is a Mendelian disorder. Physical deconditioning may exacerbate weakness; conversely, structured exercise is protective of function (see Section 12). No infectious agents are involved.

Protective factors: The single most important "protective" genetic feature is the leaky c.-32-13T>G allele itself, whose residual enzyme production yields the milder late-onset phenotype. There are no established dietary or lifestyle protective factors that prevent disease, though high-protein diet and exercise mitigate functional decline.

Gene–environment interactions: Minimal. The main modifiable interaction is treatment (ERT) and supportive exercise/nutrition acting on the genetically determined enzyme deficiency.

Section 3 — Phenotypes

LOPD is a **progressive proximal and axial myopathy with early respiratory involvement**. Key phenotypes, with suggested HPO terms and frequencies:

Phenotype	HPO term (suggested)	Type	Frequency / notes
Limb-girdle / proximal muscle weakness	HP:0003701 (proximal muscle weakness)	Clinical sign	~85% (limb-girdle phenotype 85.3%)
Respiratory / ventilatory insufficiency (diaphragm)	HP:0002093 (respiratory insufficiency); HP:0002878	Clinical sign	~61%
Axial / paraspinal weakness	HP:0003327 (axial muscle weakness)	Clinical sign	Common; early-degenerating
Elevated serum creatine kinase (hyperCKemia)	HP:0003236	Lab abnormality	Common
Exercise intolerance / fatigue	HP:0003546; HP:0012378	Symptom	Common
Minor/mild cardiac involvement	HP:0001638	Clinical sign	Minor (contrast with IOPD)
Rigid spine / scoliosis	HP:0003306; HP:0002650	Physical manifestation	Subset

The most common clinical presentation in confirmed LOPD is *"a limb-girdle phenotype (85.3%) combined with ventilatory insufficiency (61%)"* (PMID: 27170567). LOPD *"typically show[s] progressive muscle weakness, respiratory dysfunction and minor cardiac involvement"* — distinguishing it from IOPD (PMID: 31811531).

Characteristics: onset ranges from childhood to late adulthood (age-of-onset HPO: HP:0003581 adult onset); severity variable; progression slow but relentless; respiratory failure often precedes loss of ambulation.

Quality of life: LOPD *"substantially impacts patients' day-to-day activities, outcomes, and health-related quality of life"* (PMID: 39535661). Validated instruments used include PROMIS Physical Function/Fatigue, R-PAct (Pompe-specific activity), EQ-5D-5L, SF-36, and 6-minute walk distance.

Section 4 — Genetic / Molecular Information

Causal gene: *GAA* (acid alpha-glucosidase), chromosome 17q25.3, HGNC:4065, OMIM 606800. LOPD is *"a rare autosomal recessive lysosomal disorder caused by loss-of-function of the [GAA gene]"* (PMID: 40225932).

Pathogenic variants:

Feature	Detail
Most common late-onset allele	c.-32-13T>G (IVS1), splice-site, "leaky" — ~90% of Caucasian LOPD
Genetic modifier	c.510C>T (cis synonymous), lowers age of onset
Variant classes	splice-site, missense, nonsense, frameshift/indels
Pseudodeficiency alleles	c.[1726G>A;2065G>A] p.[G576S;E689K] — ~4% homozygous in East Asians
Origin	germline (biallelic)
Functional consequence	loss of function (partial residual activity in LOPD)

Variant classification follows ACMG/AMP criteria (pathogenic / likely pathogenic / VUS); ClinVar and HGMD are primary catalogs. In LOPD, at least one allele is typically the leaky c.-32-13T>G producing residual enzyme; pairing with a "null" second allele yields disease. Pseudodeficiency alleles reduce measured enzyme activity **without** causing disease and *"may present as false positives in newborn screening programs especially in Asian populations"* (PMID: 18301443).

Modifier genes/epigenetics: Beyond c.510C>T, rare variants in autophagy-related genes have been proposed as modifiers (PMID: 33807278). No established disease-specific DNA-methylation/histone signature. **Chromosomal abnormalities:** none characteristic.

Section 5 — Environmental Information

LOPD is a **monogenic Mendelian disorder with no environmental, toxic, occupational, or infectious cause**. No lifestyle factor initiates disease. Physical activity and nutrition modulate *functional outcome* rather than disease occurrence (see Section 12). No infectious agents are implicated.

Section 6 — Mechanism / Pathophysiology

Ordered causal chain

1. **Biallelic partial-LoF GAA variants** (commonly c.-32-13T>G) → **reduced synthesis/activity of acid α -glucosidase** in the lysosome (demonstrated).
2. Deficient GAA → **failure to degrade lysosomal glycogen** → **progressive intralysosomal glycogen accumulation**, first in cardiac and skeletal muscle (demonstrated).
3. Glycogen-laden lysosomes swell and **rupture**, spilling glycogen and hydrolases into the cytoplasm; concurrently an **acidification defect** develops in a subset of late endosomes/lysosomes (demonstrated in Gaa-KO myoblasts).
4. In parallel (branch), **autophagy fails at the termination stage** — impaired autophagosome–lysosome fusion → massive **autophagic buildup** restricted to **type II-rich fibers** (demonstrated).
5. Autophagic buildup **disrupts the contractile apparatus** → progressive myofiber damage and weakness; it also **acts as a sink for recombinant enzyme**, reducing ERT delivery/efficacy (demonstrated → mechanism of ERT resistance).
6. Downstream: **lipofuscin accumulation, mitochondrial dysfunction/defective oxidative phosphorylation, inflammation, apoptosis, and muscle regeneration** in vacuolated fibers (demonstrated by transcriptomics).
7. Preferential involvement of **diaphragm and axial/proximal muscles** → **respiratory insufficiency and limb-girdle weakness** → clinical LOPD, with respiratory failure as the leading cause of death (demonstrated).
8. (Branch, largely inferred for LOPD but demonstrated in models) **CNS lysosomal glycogen accumulation** occurs and is **not corrected by current ERTs** (which do not cross the blood–brain barrier).

Detail by category

Molecular pathways / cellular processes: The core lesion is a **lysosomal catabolism defect** (GO:0005764 lysosome; GO:0005980 glycogen catabolic process) with secondary **macroautophagy** dysfunction (GO:0016236; GO:0000045 autophagosome assembly; GO:0061909 autophagosome-lysosome fusion). *"The autophagic process in Pompe skeletal muscle is affected at the termination stage—impaired autophagosomal-lysosomal fusion"* (PMID: 25183957). *"Progressive accumulation of autophagic vesicles is restricted to Type II-rich muscle fibers"* and *"it also interferes with enzyme replacement therapy by acting as a sink for the recombinant enzyme and preventing its efficient delivery to the lysosomes"* (PMID: 20040311).

Protein dysfunction / biochemical abnormality: Loss of function of acid α -glucosidase (EC 3.2.1.20), a lysosomal exo-1,4- α -glucosidase. Substrate CHEBI: glycogen (CHEBI:28087); product glucose (CHEBI:17234).

Transcriptomic profiling: Single-nucleus RNA-seq plus spatial transcriptomics of LOPD muscle showed *"an increase in the proportion of slow and regenerative muscle fibres and macrophages,"* reduced glycolysis, increased lipid/amino-acid metabolism in non-vacuolated fibers (early metabolic abnormality), *"upregulation of autophagy genes and downregulation of the genes involved in ribosomal and mitochondrial function leading to defective oxidative phosphorylation,"* with inflammation/apoptosis/regeneration confined to vacuolated fibers (PMID: 39045638).

Early pathology: Even in asymptomatic patients, residual glycogen accumulates in muscle lysosomes with early glycogenin/STBD1 detection, suggesting a pre-symptomatic window and candidate biomarker (PMID: 41923452).

Immune involvement: Immunometabolic profiling of Pompe patient and mouse cells revealed *"heightened expression of activation markers in effector T cells compared to controls,"* reduced regulatory T cells, and mitophagy defects — relevant to anti-rhGAA antibody responses (PMID: 41840212).

Cell types (suggested CL terms): fast-twitch/type II skeletal myocyte (CL:0000190 striated muscle cell; CL:0002211 fast muscle cell), cardiac muscle cell (CL:0000746), macrophage (CL:0000235), motor neuron / CNS neurons (CL:0000100).

Section 7 — Anatomical Structures Affected

Organ/system level (primary): skeletal muscle (UBERON:0001134), especially proximal limb-girdle and axial/paraspinal muscles; diaphragm (UBERON:0001103) → respiratory system involvement. Secondary/minor: heart (UBERON:0000948) — minor in LOPD.

Muscle involvement pattern (imaging): *"The first muscles to degenerate in Pompe disease are hamstring and paraspinals"* with elevated glycogen (1.8–2.2× controls) preceding fatty replacement; calves are relatively spared (PMID: 41093637). Quantitative MRI identifies the adductor magnus and the age-independent adductor-magnus/rectus-femoris fat-fraction ratio as sensitive early markers (PMID: 41554264; PMID: 42223341).

Tissue/cell level: striated muscle tissue; type II (fast-twitch) myofibers preferentially; macrophage infiltration in vacuolated fibers. CNS neurons (motor neurons, anterior horn) in models.

Subcellular level (GO cellular component): lysosome (GO:0005764) — primary site of glycogen storage; autophagosome (GO:0005776); late endosome (GO:0005770) with acidification defect; mitochondria (GO:0005739) — secondary dysfunction.

Localization/lateralization: bilateral, symmetric proximal and axial muscle involvement.

Section 8 — Temporal Development

Onset: After 12 months of age by definition; ranges from childhood to the seventh–eighth decade. Insidious, chronic onset. In the Pompe Registry (n=485), *"median ages at symptom onset, diagnosis, and alglucosidase alfa initiation were 34.3, 41.1, and 44.9 years"* (PMID: 38896264) — highlighting a substantial diagnostic delay of ~7 years.

Progression: Slowly progressive and lifelong. Without ERT, forced vital capacity and ambulation decline steadily; respiratory muscle weakness drives morbidity and mortality. With alglucosidase alfa, FVC%predicted rose during the first 6 months (*"slope 1.83%/year"*) then declined (–0.54%/yr from >6 months–5 years; –1.00%/yr from >5–13 years) (PMID: 38896264). Network/observational data show initial 1–3 year gains followed by gradual 10–15 year decline (PMID: 41696977).

Critical periods: Pre-symptomatic residual-glycogen accumulation (PMID: 41923452) and subclinical qMRI changes preceding clinical decline (PMID: 42223341) define a window for early intervention. No spontaneous remission occurs.

Section 9 — Inheritance and Population

Inheritance: Autosomal recessive. Penetrance is effectively complete for biallelic pathogenic genotypes but **age of onset and expressivity are highly variable**, partly explained by the residual activity of the leaky allele and by modifier variants (c.510C>T; autophagy-gene variants). No anticipation (not a repeat-expansion disorder).

Epidemiology (birth prevalence from newborn screening): LOPD *"ranged from 1 in 82,914 in Taiwan to 1 in 17,133 in Pennsylvania"* (PMID: 40329343). In the Pennsylvania program, 531,139 newborns were screened; combined IOPD+LOPD incidence was **1:16,095** (rising to 1:8,431 including suspected LOPD) (PMID: 33202836). In Chinese newborn genomic screening, GSD-II was among the more common lysosomal disorders detected (PMID: 40355959).

Enriched clinical populations: Among adults with hyperCKemia and/or limb-girdle weakness, LOPD prevalence is **2.4% (74/3076)**, with ~95% carrying the common c.-32-13T>G splice variant (PMID: 27170567).

Carrier frequency / founder effects: GAA carriers are appreciable in the general population; in Taiwanese carrier screening, GAA was among the top autosomal-recessive genes identified, supporting reverse cascade testing (PMID: 38135707; PMID: 40673334). The c.-32-13T>G allele constitutes a Northern-European-enriched allele; pseudodeficiency alleles are enriched in East Asians (~4% homozygous) (PMID: 18301443).

Sex ratio: Approximately equal (autosomal). **Consanguinity** increases risk of biallelic disease as for any AR disorder.

Section 10 — Diagnostics

First-line enzyme test: Reduced GAA activity on a **dried blood spot (DBS)** by fluorometry or flow-injection tandem mass spectrometry (FIA-MS/MS). In Pennsylvania NBS, *"Alpha-Glucosidase (GAA) enzyme activity is measured by flow-injection tandem mass spectrometry"* with reflex full GAA sequencing ([PMID: 33202836](#)).

Confirmatory genetic test: GAA sequencing (single-gene or multigene NGS panel). Large biochemical + genetic testing programs confirm this two-step approach in >30,000 symptomatic patients ([PMID: 40225932](#)). NGS panels effectively distinguish LOPD from LGMD: *"NGS effectively diagnosed 86.2% and 13.8% of patients with LGMD and Pompe disease, respectively"* ([PMID: 39678382](#)).

Biomarkers: Urinary glucose tetrasaccharide **Glc4/Hex4** (Glcα1-6Glcα1-4Glcα1-4Glc) and serum CK. Baseline urinary Glc4 (34.2±11.3 mmol/mol creatinine) and plasma Hex4 (1.7±0.8 μM) are elevated vs controls (6.1±5.1; 0.22±0.15) and *"both urinary Glc4 and plasma Hex4 levels decreased after initiation of ERT for all patients"* ([PMID: 15886040](#)).

Imaging: Quantitative muscle MRI (Dixon fat fraction, water T2) detects subclinical involvement and disease progression before clinical decline ([PMID: 41554264](#); [PMID: 42223341](#)); 7-Tesla imaging quantifies muscle glycogen distribution ([PMID: 41093637](#)).

Electrophysiology / biopsy: EMG shows myopathic ± myotonic discharges, especially in paraspinals; muscle biopsy shows rimmed/autophagic vacuoles with PAS-positive glycogen and acid-phosphatase positivity ([PMID: 32419263](#)).

Differential diagnosis: LGMD subtypes (GNE, LDB3/ZASP, MYOT, DES, GYG1, TRIM32), Danon disease (LAMP2), inflammatory myopathies, and other late-onset vacuolar myopathies. *"An important differential diagnosis among patients presenting with proximal muscle weakness (PMW) is late-onset Pompe disease (LOPD)... which often presents with early respiratory insufficiency"* ([PMID: 31931849](#)). Danon disease (X-linked LAMP2) is a key mimic with glycogen-filled autophagic vacuoles ([PMID: 42154863](#)).

Screening: Newborn screening (DBS enzyme + reflex sequencing); carrier and reverse cascade testing in families ([PMID: 40673334](#)). Pseudodeficiency alleles must be recognized to avoid false positives ([PMID: 18301443](#)).

Section 11 — Outcome / Prognosis

Mortality: Respiratory failure from progressive diaphragmatic weakness is the leading cause of death. ERT substantially improves survival: in a Chinese single-center cohort (n=68), *"the mortality rate was 6.4%, compared to 57.1% in the 21 patients who did not receive ERT"* (PMID: 42131243).

Morbidity / function: Progressive loss of ambulation and respiratory capacity; many patients ultimately require ventilatory support and mobility aids. Quality-of-life burden is substantial across PROMIS, R-PAct, EQ-5D-5L, and SF-36 measures (PMID: 39535661; PMID: 39318718).

Disease course with treatment: ERT changes the natural history but effectiveness wanes over time; initial 1–3 year gains are followed by gradual decline over 10–15 years (PMID: 41696977). A proposed "therapeutic corridor of stability" defines FVC change of approximately –1% to +5%/year and 6MWD stability within ± 25 m of peak as adequate response (PMID: 42538546).

Prognostic factors/biomarkers: Age at ERT initiation, baseline muscle involvement (qMRI fat fraction), FVC, 6MWD, and biomarker trajectory (Glc4/Hex4, CK). Persistently elevated Glc4 tracks limited motor improvement (PMID: 15886040).

Section 12 — Treatment

Enzyme replacement therapy (ERT) — mainstay

Three approved agents (suggested NCIT: Enzyme Replacement Therapy):

Agent	Key feature	Pivotal trial
Alglucosidase alfa	Standard of care >15 years	LOTS
Avalglucosidase alfa	Enhanced bis-M6P → improved muscle uptake	COMET (NCT02782741)
Cipaglucosidase alfa + miglustat	Enzyme + chaperone stabilizer	PROPEL (NCT03729362)

Alglucosidase alfa *"has been available for more than 15 years and is the standard treatment... although the effectiveness of the treatment reduces over time"* (PMID: 41453611).

Comparative efficacy: In network meta-analysis, ERT-naïve patients showed *"significant 6MWD improvements vs. placebo: ~25 m with alglucosidase alfa and ~54 m with avalglucosidase alfa"* at ~1 year (PMID: 40842017; PMID: 41696977). Real-world data confirm *"avalglucosidase alfa and cipaglucosidase alfa plus miglustat have demonstrated that both treatments are at least as efficacious as alglucosidase alfa"* (PMID: 40471681). Indirect comparisons numerically favor avalglucosidase over cipa+mig for FVC and 6MWT (PMID: 40920287; PMID: 41757410).

Switching: ERT-experienced patients switched from alglucosidase to cipaglucosidase alfa+miglustat improved or stabilized across most outcomes, with SGIC responder rates *"90 vs. 59% responders in the cipa+mig vs. the alg+pbo group"* (PMID: 39535661; PMID: 40342075). Switching to avalglucosidase is likewise feasible and safe (PMID: 38313679).

Advanced therapeutics

Gene therapy: Systemic AAV9-GAA is in development, including novel liver/muscle-restricted promoters (G6PC, CRM4-G6PC) that evade transgene-specific immune responses in mice (PMID: 41485391). However, safety risks exist: *"a late onset Pompe disease patient who developed sensory neuropathy after initiating gene therapy with AT845"* (DRG toxicity) (PMID: 42142433).

Supportive / rehabilitative (adjuvant to ERT)

Supervised **resistance training (24 weeks) plus respiratory-muscle training** improved leg extensor strength ($+10.5 \pm 3.2$ Nm, $p < 0.01$), leg flexors ($+12.1 \pm 4.1$ Nm, $p < 0.01$) and maximal inspiratory pressure ($+8.5 \pm 3.7$ cmH₂O) in ERT-treated LOPD patients (PMID: 35365393). **High-protein diet plus physical training** improved muscle strength, fatigue, and physical QoL in pediatric Pompe patients (PMID: 37002894). Structured adapted physical activity and pulmonary rehabilitation are recommended (PMID: 31811531; PMID: 31970320).

Immunogenicity considerations

Anti-rhGAA antibodies can reduce ERT efficacy. *"ERT response depends on several factors, including ERT initiation age, dose, and cross-reactive immunological material (CRIM) status, especially in infantile-onset Pompe disease"* (PMID: 41583481). Most LOPD patients carry the leaky allele producing some GAA protein (CRIM-positive) and develop lower titers, but immunomodulation (rituximab + methotrexate + IVIG) is critical mainly for CRIM-negative IOPD, where *"the inability of CRIM-negative IOPD patients to produce native GAA exposes them to a high risk of development of anti-rhGAA IgG antibody titers, leading to treatment failure"* (PMID: 39314994).

Section 13 — Prevention

There is no primary prevention for this Mendelian disorder. **Secondary prevention** relies on early detection: **newborn screening** (DBS GAA enzyme + reflex sequencing) enables presymptomatic identification and timely ERT ([PMID: 33202836](#); [PMID: 40355959](#)), and **carrier / reverse cascade testing** identifies at-risk relatives ([PMID: 40673334](#); [PMID: 38135707](#)). **Genetic counseling**, carrier screening, prenatal testing, and preimplantation genetic diagnosis inform reproductive decisions. **Tertiary prevention** (preventing complications) comprises ERT, respiratory support, vaccination against respiratory infections, exercise/nutrition, and multidisciplinary follow-up per LOPD management recommendations ([PMID: 41453611](#)).

Section 14 — Other Species / Natural Disease

Naturally occurring GAA/Pompe disease ("generalized glycogenosis") is well documented in animals. *"Generalized glycogenosis is a lethal autosomal recessive disease caused by a deficient activity of the acidic 1,4- α -glucosidase enzyme and characterized by an accumulation of glycogen within lysosomes"* in cattle ([PMID: 31377960](#)).

Cattle (*Bos taurus*, NCBI:txid9913): Loss-of-function GAA alleles are established in **Brahman** (E13, 1783C>T), **Shorthorn** (E18, 2454delCA), and **Droughtmaster/Braford** (E7, 1057delTA), all causing premature translation termination; carrier frequency ~12% in Argentinean Brahman-derived herds. Homozygous calves show multisystem pathology: *"Pathology consistent with generalised glycogenosis was found in the skeletal and cardiac muscle and spinal cord of both of the affected calves"* ([PMID: 28444756](#)). Bovine GAA is *"highly conserved compared with the human alpha-glucosidase gene (86% and 83% identity respectively)"* ([PMID: 10723725](#)). Heterozygote detection can be confounded by haemopoietic chimaerism in twin cattle ([PMID: 8161014](#)).

Other species: Naturally occurring acid maltase deficiency is reported in **dogs** (e.g., Lapland dog), **cats**, **sheep**, and **Japanese quail** — the last historically used as an avian therapy-development model. **Orthologous gene:** GAA is conserved; suggested resources: OMIA, NCBI Gene, HomoloGene.

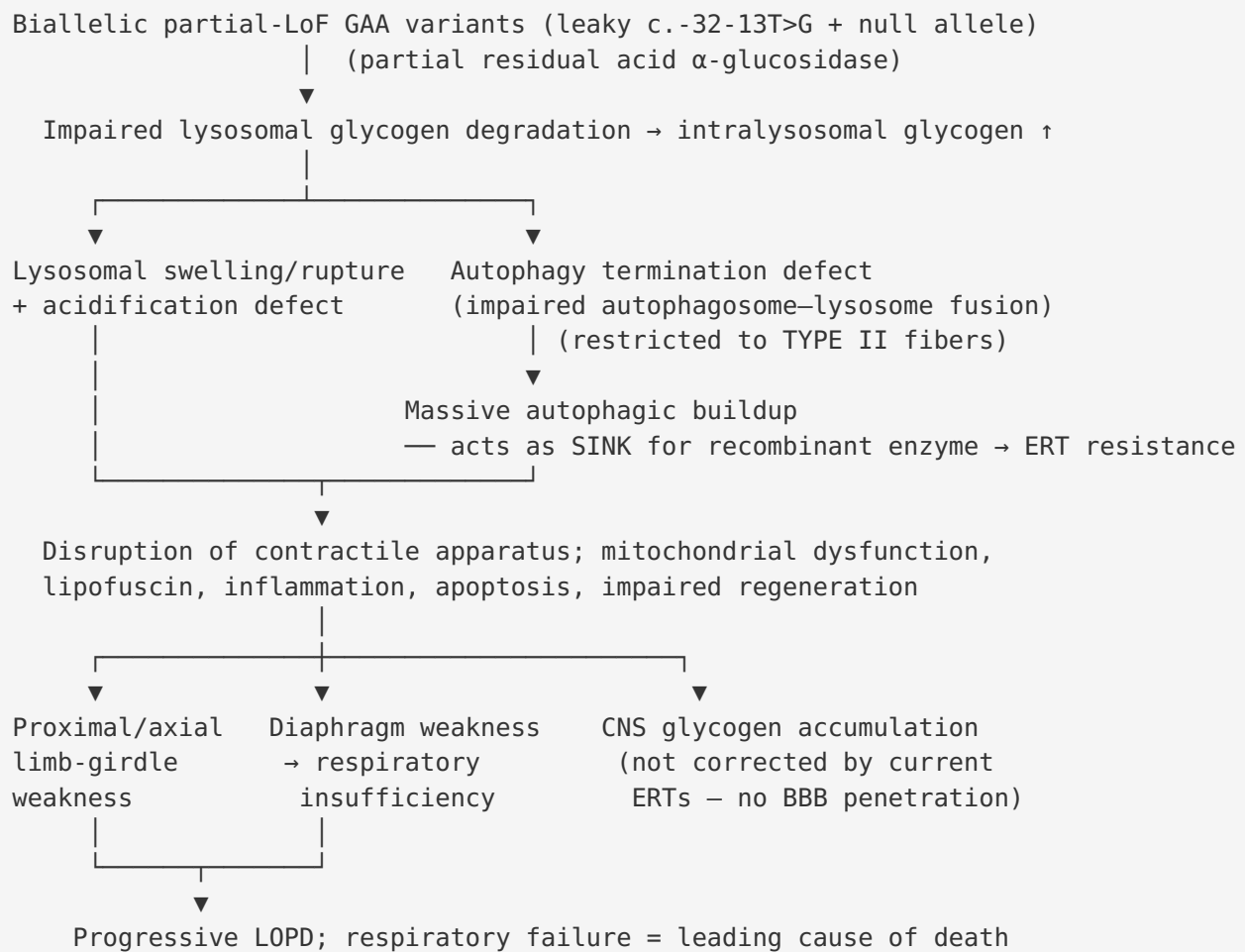
Section 15 — Model Organisms

Mammalian genetic models: The **Gaa-knockout mouse** is the principal model; newer CRISPR knock-in / compound-heterozygous mice recapitulate the metabolic hallmark. *"CHet mice exhibited reduced GAA activity and elevated plasma glycogen levels, recapitulating the metabolic hallmark of human PD"* (PMID: 41742217). Models reproduce cardiac, skeletal, diaphragmatic and hepatic glycogenosis with **type II fiber-restricted autophagic buildup** and an endosomal/lysosomal acidification defect (PMID: 16532490; PMID: 20040311).

Applications & extensions: - **CNS involvement:** *"none of the approved therapies are known to cross the blood brain barrier (BBB) and patients with PD have progressive central nervous system (CNS)-associated impairments due to lysosomal glycogen accumulation in the CNS"* — motivating focused-ultrasound BBB opening for ERT delivery (PMID: 41349290). - **Gene therapy:** AAV9-GAA with novel promoters evading immune responses (PMID: 41485391). - **Immunometabolism:** heightened effector T-cell activation and mitophagy defects in Gaa-/- immune cells (PMID: 41840212).

Limitations: Mouse models show variable fidelity to the slowly progressive, respiratory-predominant adult human phenotype; the human leaky splice allele (a splicing phenomenon) is difficult to model directly. Satellite-cell dysfunction may contribute to impaired regeneration across neuromuscular disorders including Pompe (PMID: 35302338). **Databases:** MGI, IMPC, IMSR; cellular models include patient iPSC-derived myotubes.

Mechanistic Model / Interpretation



The unifying insight is that LOPD is **not simply a lysosomal storage problem but a disorder of failed autophagy in a specific fiber type**. The type II-fiber-restricted autophagic buildup both damages muscle and blunts ERT delivery, explaining why enzyme therapy modifies without curing the disease and why next-generation, higher-uptake enzymes and adjuvant strategies matter.

Evidence Base — Key Literature

PMID	Contribution	Relationship to findings
30922962	c.-32-13T>G in ~90% of Caucasian LOPD; c.510C>T modifier	Supports F001 (genetics)
20040311	Type II-restricted autophagic buildup; ERT "sink"	Supports F002 (mechanism)
25183957	Impaired autophagosome-lysosome fusion	Supports F002
16532490	Endocytic/autophagic dysfunction, acidification defect	Supports F002
40329343	NBS birth prevalence range	Supports F003 (epidemiology)
27170567	2.4% LOPD in hyperCKemia/LGMW; phenotype frequencies	Supports F003, F004
18301443	Pseudodeficiency alleles → NBS false positives	Supports F003
38896264	Registry timeline; FVC trajectory on ERT	Supports F004
42131243	Mortality 6.4% (ERT) vs 57.1% (no ERT)	Supports F004
40842017 / 41696977	ERT 6MWD network meta-analysis	Supports F005
40471681	Next-gen ERTs ≥ alglucosidase	Supports F005
42142433	Gene therapy AT845 sensory neuronopathy	Supports F005 (risk)
41742217	Compound-het mouse recapitulates hallmark	Supports F006
41349290	CNS glycogen; ERT no BBB penetration	Supports F006
41840212	Immunometabolic dysregulation	Supports F006
39535661	Multisystem/QoL; PRO on switching	Supports F007
39678382 / 31931849 / 32419263	LGMD/vacuolar-myopathy differential; NGS	Supports F008
41583481 / 39314994	CRIM/immunogenicity	Supports F009

PMID	Contribution	Relationship to findings
28444756 / 31377960 / 10723725	Bovine natural disease; GAA conservation	Supports F010
31811531 / 35365393 / 37002894	Rehab/nutrition adjuncts; minor cardiac	Supports F011
15886040	Glc4/Hex4 biomarker	Supports F012

Limitations and Knowledge Gaps

- **Prevalence estimates** vary widely by ascertainment method (newborn screening vs. clinical cohorts) and geography (1:17,133 to 1:82,914 for LOPD from NBS); true adult clinical prevalence is uncertain because of diagnostic delay (~7 years).
- **Genotype–phenotype correlation is imperfect:** patients with identical GAA genotypes vary in onset and severity, implicating incompletely characterized modifier genes and epigenetic factors ([PMID: 33807278](#)).
- **Comparative ERT efficacy** rests largely on indirect/network comparisons rather than head-to-head trials; long-term (10–15 year) comparative outcomes for next-generation agents are immature ([PMID: 41696977](#)).
- **CNS involvement in human LOPD** is inferred largely from models; its clinical significance and the value of BBB-crossing therapies remain to be established.
- **Gene therapy** safety (DRG toxicity) requires resolution before broad application.
- **Biomarkers** (Glc4/Hex4, qMRI) need longitudinal validation as surrogate endpoints and for optimal timing of therapy initiation.

Proposed Follow-up Experiments / Actions

1. **Longitudinal qMRI + biomarker validation:** Prospectively validate adductor-magnus fat fraction and urinary Glc4 as surrogate endpoints and triggers for pre-symptomatic ERT initiation.

2. **Modifier-gene discovery:** Genome-wide/whole-genome sequencing of discordant siblings sharing GAA genotype to identify severity modifiers beyond c.510C>T and autophagy-gene variants.
3. **Head-to-head next-generation ERT trial:** Randomized comparison of avalglucosidase alfa vs. cipaglucosidase alfa+miglustat with harmonized endpoints (FVC, 6MWD, Glc4, PROs).
4. **CNS-directed therapy translation:** Advance focused-ultrasound BBB-opening and CNS-tropic AAV-GAA approaches, with rigorous DRG-toxicity monitoring informed by the AT845 signal.
5. **Autophagy-targeted adjuvant strategies:** Test agents that restore autophagosome-lysosome fusion to relieve the "enzyme sink" and improve ERT delivery to type II fibers.
6. **Standardized exercise/nutrition protocols:** Multi-center RCT of resistance + respiratory-muscle training plus high-protein diet as adjuvant to next-generation ERT.
7. **Reverse cascade testing implementation:** Operationalize family-wide screening following NBS-positive LOPD to shorten diagnostic delay in relatives.

Evidence source types: findings are drawn from human clinical cohorts and registries, randomized controlled trials, newborn-screening programs, model-organism (mouse, cattle) studies, in vitro/muscle-biopsy transcriptomics, and quantitative imaging. All mechanistic and clinical claims are cited to primary literature by PMID.
